

DekantPM

Mathematical Analysis

L²-Norm AMM for Continuous Prediction Markets
Bin Discretization • Distribution Trading • Solvency

Solana • Anchor 0.32 • u128 Fixed-Point Arithmetic

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1. Introduction: Continuous Prediction Markets

1.1 What Are Prediction Markets?

A prediction market is a speculative market where participants trade contracts whose payoffs depend on the outcome of future events. In a binary prediction market, a contract pays \$1 if event E occurs and \$0 otherwise. The market price reflects the crowd's aggregate belief about the probability of E.

1.2 From Discrete to Continuous

Traditional prediction markets handle discrete outcomes. But many real-world quantities are continuous: temperatures, prices, dates. A continuous prediction market elicits a full probability distribution over a continuous outcome space — the crowd's aggregate PDF over all possible answers.

1.3 The Goal

1. Aggregate beliefs — Allow participants to express nuanced probabilistic views.
2. Maintain a consensus distribution — The market state encodes the crowd's aggregate PDF.
3. Incentivize accuracy — Traders who move the distribution toward truth profit.
4. Provide liquidity — An AMM ensures trades can always execute without counterparties.

2. Mathematical Framework: The L₂-Norm AMM

2.1 Core Setup

Consider a market with N outcomes (or bins). We define:

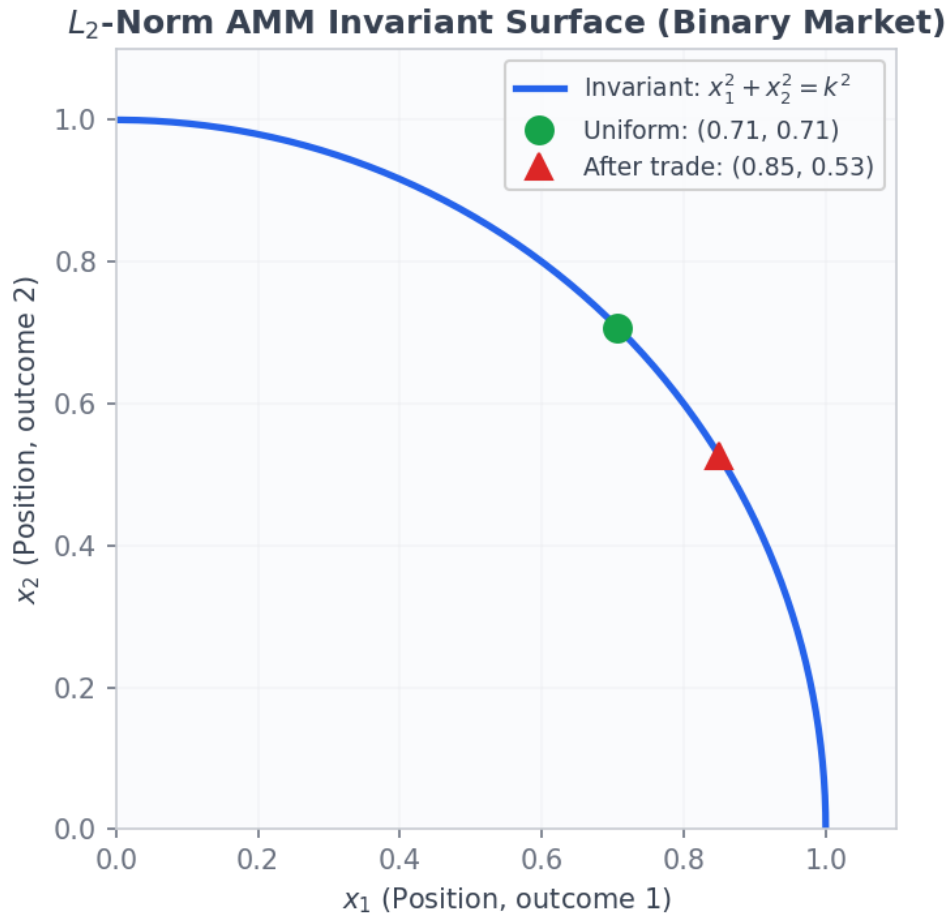
- k (= total_minted): Total collateral deposited as complete sets.
- $h = (h_1, \dots, h_n)$: AMM reserves per outcome.
- $x = (x_1, \dots, x_n)$: Positions held by traders, where $x_i = k - h_i$.

2.2 The Invariant

The AMM enforces a constant L₂-norm on positions:

$$\|x\|_2 = \sqrt{\sum_{i=1}^N x_i^2} = k \iff \sum_{i=1}^N x_i^2 = k^2$$

This defines a hypersphere of radius k in position space. All valid AMM states lie on this surface.

Figure 1: L_2 -norm invariant surface for a binary market.

2.3 Why L_2 -Norm? — Market Scoring Rule

Suppose the true distribution is $\mathbf{p} = (p_1, \dots, p_n)$. A rational trader maximizes expected value on the invariant surface:

$$\max_{\mathbf{x}} \mathbf{p} \cdot \mathbf{x} \quad \text{s.t.} \quad \|\mathbf{x}\|_2 = k$$

By Cauchy-Schwarz, the optimum is:

$$\mathbf{x}^* = k \cdot \frac{\mathbf{p}}{\|\mathbf{p}\|_2}$$

So at equilibrium $x_i \propto p_i$ — positions directly encode the probability distribution.

2.4 Probability Extraction

Linear extraction (true equilibrium probabilities):

$$p_i = \frac{x_i}{\sum_j x_j}$$

Quadratic extraction (used by DekantPM):

$$\hat{p}_i = \frac{x_i^2}{k^2}$$

The quadratic values sum to 1 by the invariant. At equilibrium:

$$\hat{p}_i = \frac{p_i^2}{\|\mathbf{p}\|_2^2}$$

The implementation uses fixed-point scaling with $S = 10^9$:

$$\hat{p}_{i, \text{scaled}} = \frac{x_i^2 \cdot S}{k^2}$$

2.5 Quadratic Pricing: Distortion Analysis

The quadratic formula is NOT the true probability — it is the squared and renormalized probability. This compresses values toward the extremes:

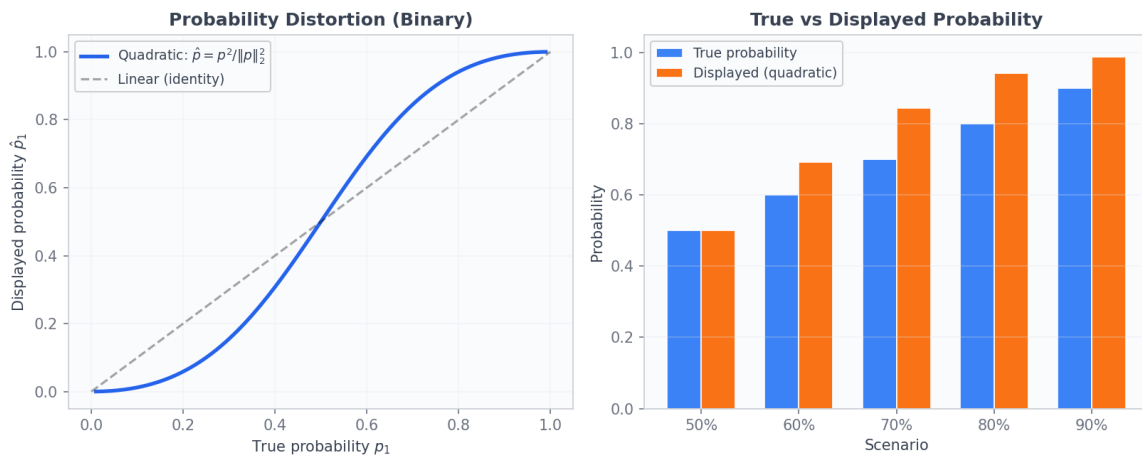


Figure 2: Quadratic distortion. True 70% displays as 84.5% in a binary market.

To recover true probabilities from displayed values:

$$p_i = \frac{\sqrt{\hat{p}_i}}{\sum_j \sqrt{\hat{p}_j}}$$

The quadratic formula is used because: (1) it sums to 1 by the invariant, (2) all AMM operations use it consistently, (3) outcome ordering is preserved.

Convention

Throughout this document, $\hat{p}_i$ refers to the quadratic extraction x_i^2/k^2 unless otherwise noted.

3. Market Initialization

With initial liquidity L and N outcomes:

$$k = L, \quad x_i = \lfloor \sqrt{L^2/N} \rfloor, \quad h_i = L - x_i, \quad k^2 = L^2$$

Initial probability: $\hat{p}_i = x_i^2/k^2 \approx 1/N$ (uniform).

Example (binary, $L = 1,000,000$):

$$\begin{aligned} x_1 &= x_2 = 707,106 \\ h_1 &= h_2 = 292,894 \\ \hat{p} &= 0.5 \end{aligned}$$

4. Discrete Trading

4.1 Buy (Single Outcome)

Trader spends c collateral (after fees) to buy tokens of outcome i :

1. Mint complete sets: $h_j \leftarrow h_j + c$, $k' = k + c$.
2. Positions unchanged after mint: $x'_j = x_j$ for all j .
3. Solve for new position of target outcome:

$$x'_i = \sqrt{k'^2 - \sum_{j \neq i} x_j^2}$$

4. Tokens out = $x'_i - x_i$. Update $h_i = k' - x'_i$.

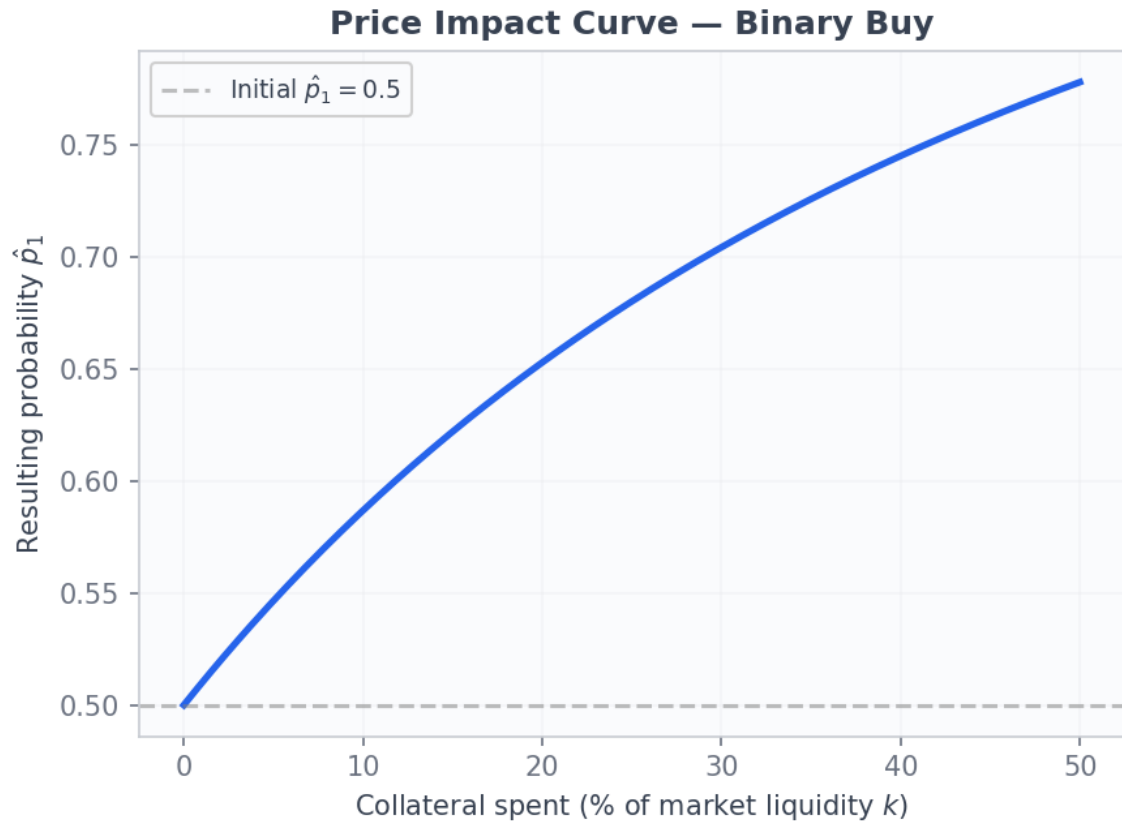


Figure 3: Price impact — concave relationship provides natural slippage protection.

4.2 Sell (Single Outcome)

Trader returns t tokens of outcome i :

1. Return: $h_i \leftarrow h_i + t$, so $x'_i = x_i - t$.
2. New total minted:

$$k' = \sqrt{\sum_j x_j'^2}$$

Collateral out = $k - k'$.

3. Burn complete sets: $h_j \leftarrow h_j - (k - k')$ for all j .

4.3 Price Impact

For a buy of amount c on outcome i , the price impact increases with c/k (trade size relative to liquidity). For small trades ($c \ll k$), impact is approximately linear; for large trades, the square root creates diminishing returns — each additional unit of collateral buys fewer tokens.

5. Distribution Trading (Continuous Markets)

5.1 Bin Discretization

DekantPM discretizes $[a, b]$ into N bins. Bin j has center:

$$\text{center}_j = a + \frac{(2j+1)(b-a)}{2N}$$

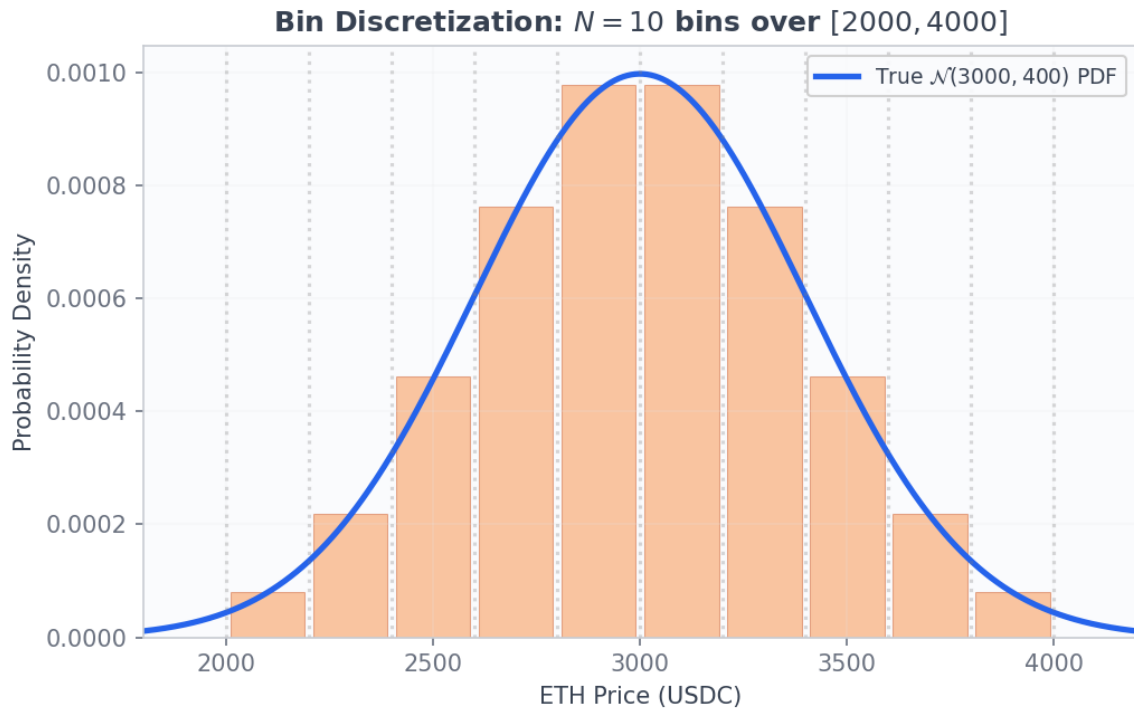


Figure 4: Bin discretization approximates the continuous Normal PDF as a histogram.

5.2 Normal PDF Bin Weights

For each bin j , compute the standardized score:

$$z_j = \frac{\text{center}_j - \mu}{\sigma}$$

If $|z_j| > 5$, weight = 0. Otherwise, approximate via degree-4 Taylor polynomial:

$$w_j = e^{-z_j^2/2} \approx 1 - \frac{t}{2} + \frac{t^2}{8} - \frac{t^3}{48} + \frac{t^4}{384} \quad (t = z^2)$$

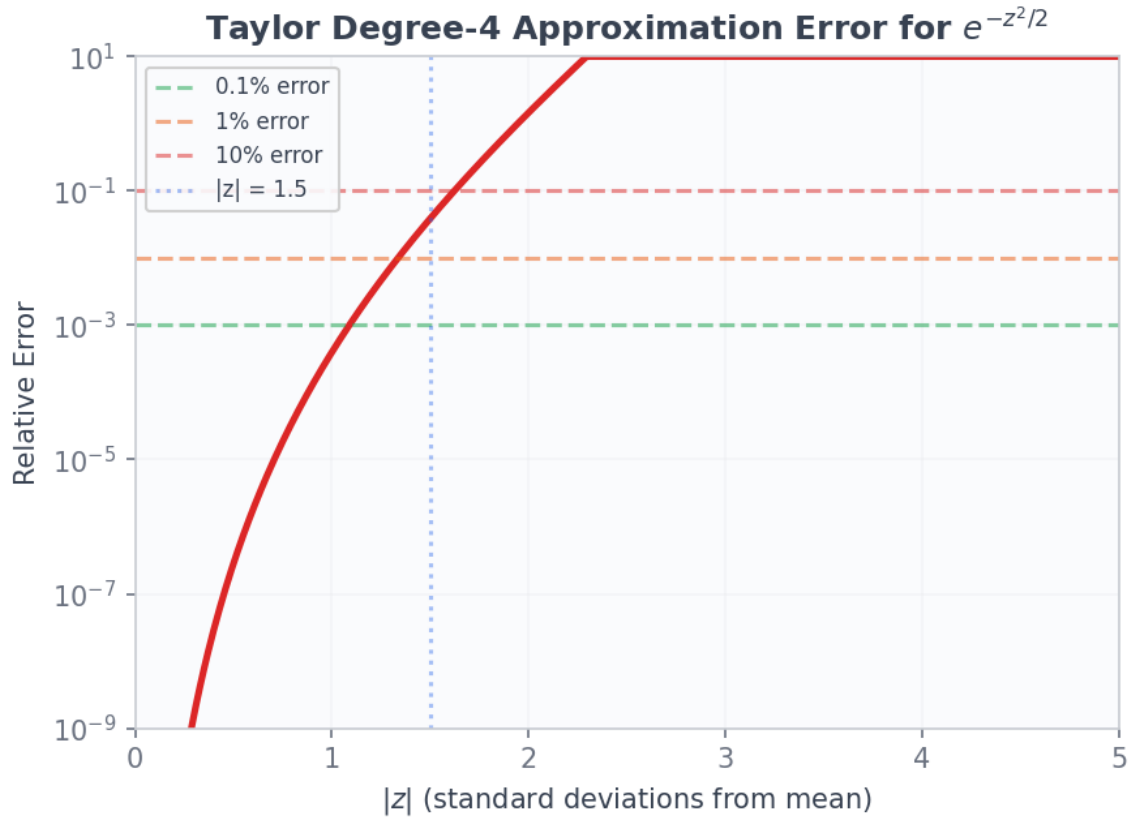


Figure 5: Taylor approximation error — excellent for $|z| < 1.5$, acceptable to $|z| \approx 3$.

Weights are normalized so they sum to $S = 10^9$:

$$W_j = \frac{w_j \cdot S}{\sum_m w_m}, \quad \sum_j W_j = S$$

5.3 Distribution Buy

Trader spends c to buy a weighted bundle. Define intermediate sums:

$$XW = \sum_j x_j W_j, \quad W^2 = \sum_j W_j^2$$

Then the scaling parameter is:

$$\lambda = \sqrt{XW^2 + W^2 \cdot \text{excess}} - XW \quad (\text{excess} = k'^2 - k^2)$$

And each bin receives:

$$\text{tokens_out}_j = \frac{\lambda \cdot W_j}{W^2}$$

5.4 Distribution Sell

Return T tokens proportional to W . Compute:

$$\begin{aligned} x'_j &= x_j \cdot \frac{T \cdot W_j}{S} \\ k' &= \sqrt{\sum_j x_j'^2} \\ \text{collateral_out} &= k - k' \end{aligned}$$

5.5 Buy-to-Price

Target $\hat{p}_i = p^*/S$ for outcome i :

$$\text{collateral} = \sqrt{\frac{\sum_{-i} \cdot S}{S - p^*}} - k$$

where $\sum_{-i} = \sum_{\{j \neq i\}} x_j^2$.

6. Liquidity Provision

6.1 Adding Liquidity

LP deposits d . Reserves scale proportionally:

$$h'_j = h_j \cdot \frac{k+d}{k}$$

Probabilities are preserved. First LP: shares = L . Subsequent:

$$\text{shares} = L_{\text{total}} \cdot \frac{d}{k}$$

6.2 Removing Liquidity

Active market:

$$\text{collateral_out} = k \cdot \frac{S}{L_{\text{total}}}$$

Resolved market:

$$\text{collateral_out} = h_{\text{winning}} \cdot \frac{S}{L_{\text{total}}}$$

Plus fee share in both cases.

6.3 LP Economics

LPs earn fees (0.15%/trade default) and profit from mean reversion. They face divergence loss and adverse selection from informed traders.

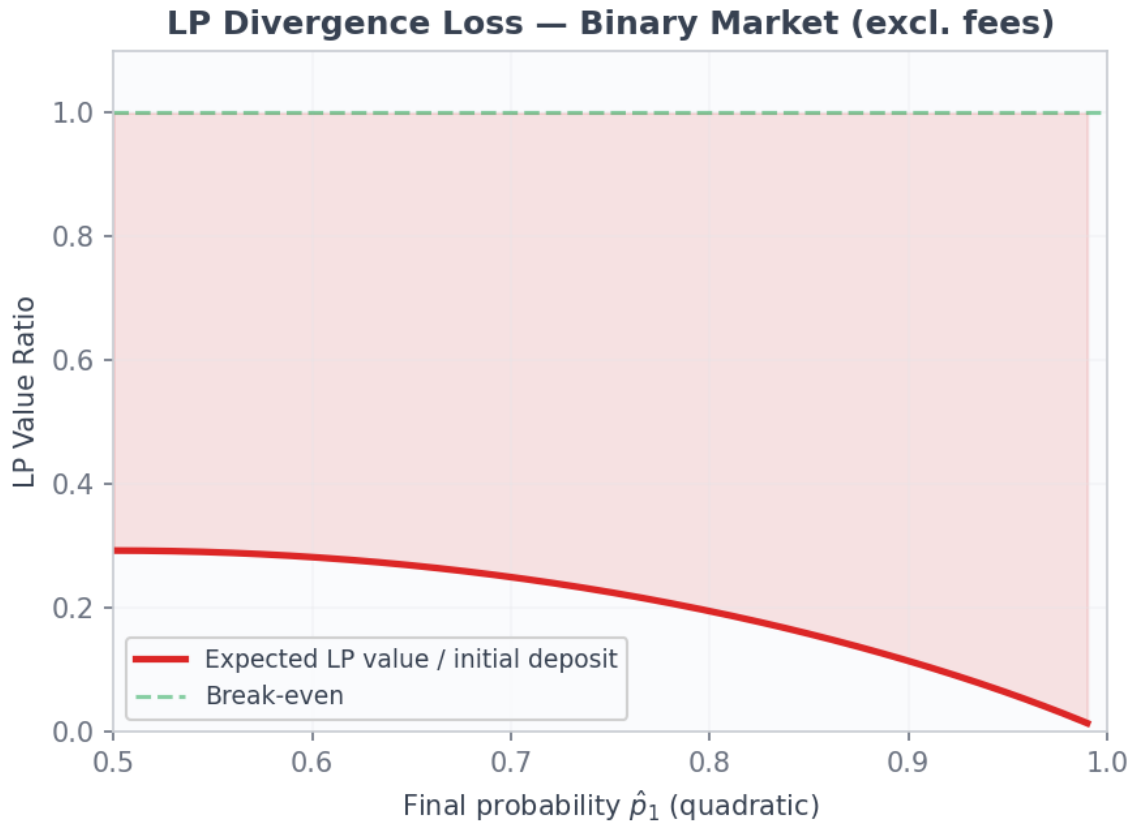


Figure 6: LP divergence loss in a binary market (excluding fees).

7. Resolution and Payout

7.1 Discrete Markets

Oracle provides winning outcome. Winners redeem 1:1 minus redemption fee (default 0.5%).

7.2 Continuous Markets

Oracle provides value $v \in [a, b]$. Winning bin is:

$$\text{bin} = \left\lfloor \frac{(v-a) \cdot N}{b-a} \right\rfloor, \quad \text{clamped to } [0, N-1]$$

7.3 Solvency Guarantee

The vault always holds $\geq k$ collateral. Maximum redemption:

$$\max(x_i) \leq k \quad \text{since} \quad x_i^2 \leq \sum_j x_j^2 = k^2 \quad \blacksquare$$

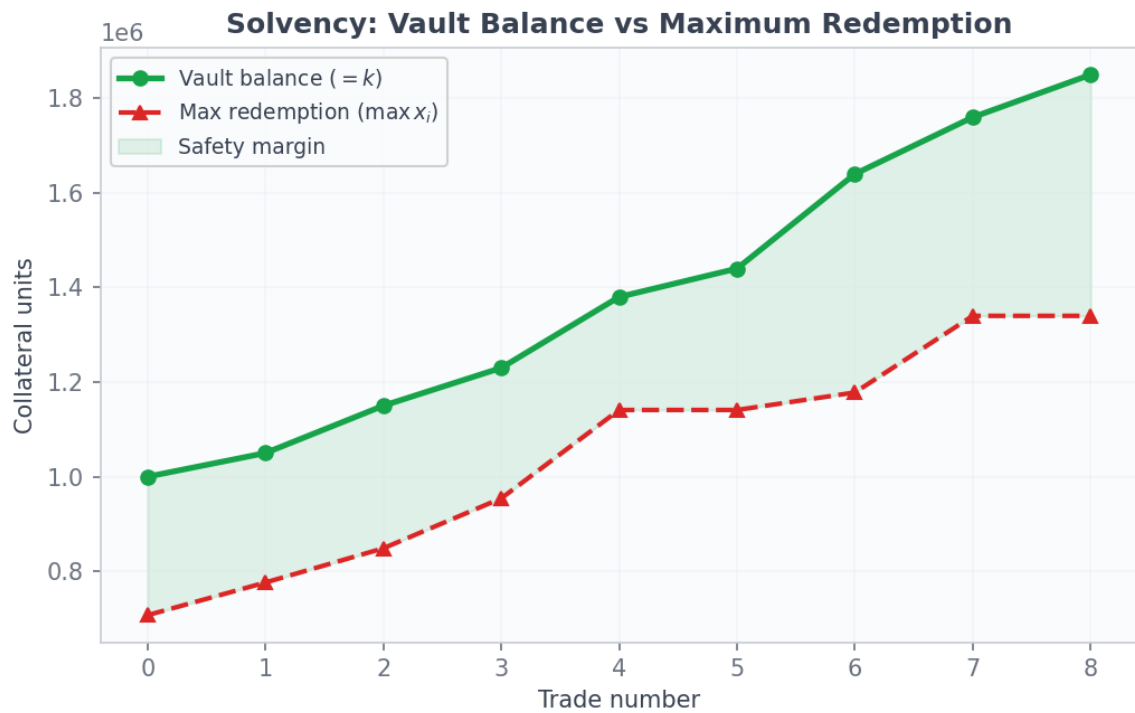


Figure 7: Solvency — vault balance always exceeds maximum possible redemption.

8. Worked Example: Continuous Market Lifecycle

8.1 Market Creation

Market: "ETH price March 31." Range [2000, 4000], N=5 bins, L = 1,000,000.

Bin	Range	Center
0	[2000, 2400)	2200
1	[2400, 2800)	2600
2	[2800, 3200)	3000
3	[3200, 3600)	3400
4	[3600, 4000]	3800

$$k = 1,000,000, \quad x_i = 447,213, \quad h_i = 552,787, \quad \hat{p}_i = 0.200$$

8.2 Trade 1 — Alice Buys Bin 2

Alice buys bin 2 with 100,000 collateral. Fee = 300, net = 99,700.

$$\begin{aligned}
 k' &= 1,099,700 \\
 \Sigma_{-2} &= 4 \times 447,213^2 = 799,997,549,476 \\
 x'_2 &= \sqrt{k'^2 - \Sigma_{-2}} = 639,798 \\
 \text{tokens_out} &= 192,585, \quad \hat{p}_2 = 0.339
 \end{aligned}$$

8.3 Trade 2 — Bob's Distribution Buy $N(3000, 400)$

Bob spends 200,000 expressing $N(\mu=3000, \sigma=400)$. Bin weights concentrate on bins 1-3:

Bin	z	$\exp(-z^2/2)$	Weight ($\times 10^9$)
0	-2.0	0.135	54.5M
1	-1.0	0.607	244.3M
2	0.0	1.000	402.5M
3	+1.0	0.607	244.3M
4	+2.0	0.135	54.3M

After trade: bell-shaped distribution, bin 2 at ~40%.

8.4 Liquidity + Resolution

Carol adds 500,000 LP — depth increases, probabilities unchanged. Oracle reports ETH = \$3,150.

$$\text{bin} = \left\lfloor \frac{(3150 - 2000) \times 5}{2000} \right\rfloor = 2$$

Alice's 192,585 tokens → net 191,623 (91.6% profit). Bob profits on bin-2 tokens only. Carol gets LP residual + fees.

Continuous Market Lifecycle — Probability Evolution

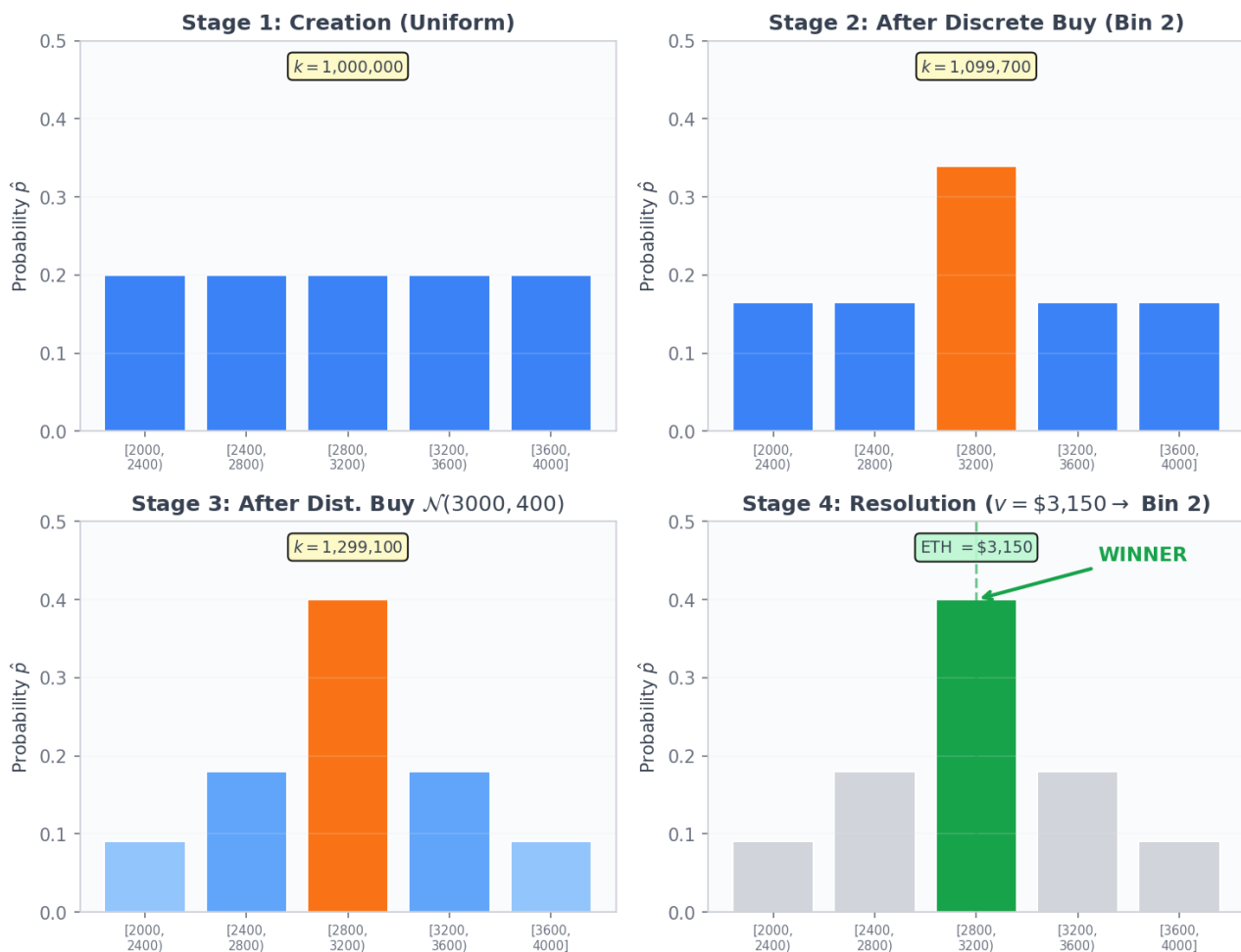


Figure 8: Probability evolution through the market lifecycle.

9. Fee Structure

On every trade with gross amount G , fees are split between LPs and the protocol:

$$\begin{aligned} \text{total_fee} &= \left\lfloor \frac{G \times \text{trade_fee_bps}}{10,000} \right\rfloor \quad (\text{default } 30 \text{ bps} = 0.3\%) \\ \text{lp_fee} &= \left\lfloor \frac{\text{total_fee} \times \text{lp_share_bps}}{10,000} \right\rfloor \quad (\text{default } 50\%) \\ \text{protocol_fee} &= \text{total_fee} - \text{lp_fee} \\ \text{redemption_fee} &= \left\lfloor \frac{\text{payout} \times \text{redemption_bps}}{10,000} \right\rfloor \quad (\text{default } 50 \text{ bps}) \end{aligned}$$

A round-trip (buy then sell) always costs at least $2 \times \text{fee_rate}$, creating a natural bid-ask spread. LP fees are accumulated on the market account and distributed proportionally when LPs withdraw.

10. The Discretization Approach

10.1 Advantages

1. Simplicity — Same AMM engine for discrete and continuous markets.
2. On-chain efficiency — $O(N)$ operations, max $N=256$.
3. Composability — Individual bin tokens can be traded separately.
4. Exact solvency — Complete-set minting guarantees solvency.
5. No backing constraint — Unlike Paradigm's $\max(f) \leq b$.

10.2 Disadvantages

1. Resolution granularity — Same-bin values treated identically.
2. Approximation error — PDF becomes step function.
3. Fixed bins — Equal spacing may waste resolution.
4. Taylor error — Degrades for $|z| > 1.5$.
5. Normal-only — Cannot express skew or multimodality.

11. Limitations and Alternative Approaches

11.1 Single-Bin Resolution

Problem: Bin boundary creates payout discontinuity. Alternative: Pro-rata resolution distributes payouts to nearby bins proportional to distance from resolved value.

11.2 Fixed Number of Bins

Problem: N fixed at creation; consensus narrowing wastes bins. Alternative: Adaptive binning or hierarchical child markets.

11.3 Normal-Only Distributions

Problem: Only $N(\mu,\sigma)$ expressible. Alternative: Support log-normal, uniform, mixtures, or arbitrary weight vectors.

11.4 Taylor Polynomial Precision

Problem: The degree-4 Taylor approximation of $\exp(-z^2/2)$ degrades for $|z| > 1.5$. For narrow distributions where bins at $|z| \approx 2$ -3 carry meaningful weight, errors propagate. Alternative: Degree-8 or piecewise polynomial approximation.

11.5 Compute Budget

Problem: Distribution buy/sell iterates over all N bins. For $N=256$, this approaches Solana's compute unit limits. Alternative: Batch processing across multiple transactions, or an AMM operating on distribution parameters directly.

11.6 Quadratic Probability Distortion

Problem: $\hat{p}_i = x_i^2/k^2$ distorts true probabilities (Section 2.5). Alternative: Display linear probabilities $x_i/\sum x_j$ alongside quadratic market prices.

12. Comparison with Paradigm's Distribution Markets

12.1 Overview

Paradigm's paper (Dave White, Dec 2024) proposes continuous prediction markets in infinite-dimensional L^2 space. Traders hold payout functions $f:\mathbb{R}\rightarrow\mathbb{R}_+$, AMM enforces $\|f\|_2 = k$, solvency requires $\max(f) \leq b$.

12.2 Structural Comparison

Aspect	DekantPM	Paradigm
Outcome space	Finite: N bins	Infinite: $\mathbb{R}$
Invariant	$\sum x_i^2 = k^2$	$\int f^2 dx = k^2$
Solvency	Automatic	$\max(f) \leq b$
Resolution	Step function	Smooth: $f(v)$
Collateral	Complete-set mint	Max-loss based
On-chain	$O(N)$ per trade	$O(1)$ + numerics

12.3 Backing Constraint

In finite dimensions:

$$\|\mathbf{x}\|_2 = k \implies |x_i| \leq k$$

In infinite dimensions, $\|f\|_2 = k$ does NOT bound $\max(f)$. Hence Paradigm needs:

$$\max(f) \leq b \quad \text{and} \quad \sigma \geq \frac{k^2}{b^2 \sqrt{\pi}}$$

DekantPM sidesteps this via complete-set minting.

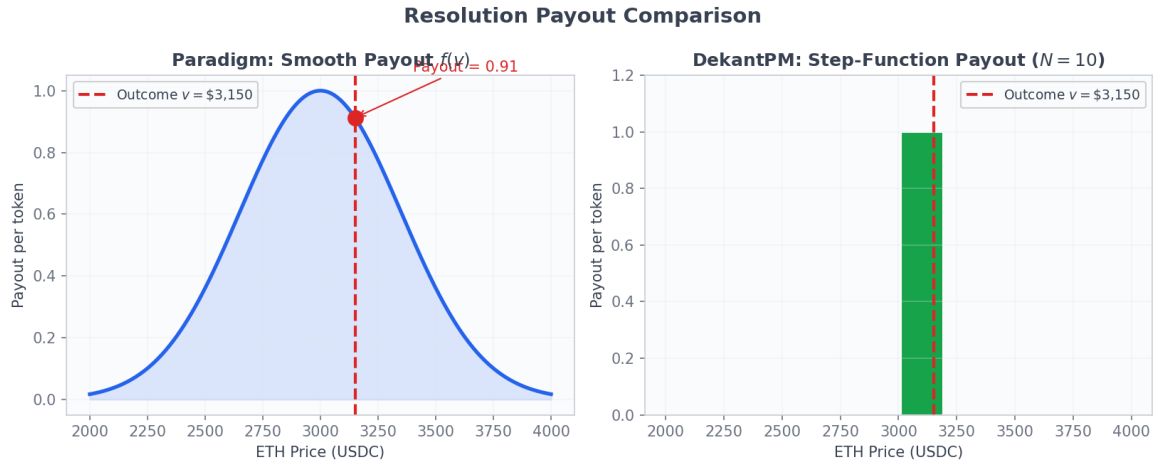


Figure 9: Paradigm gives smooth payouts; DekantPM gives winner-take-all per bin.

12.4 Convergence

As $N \rightarrow \infty$, DekantPM converges to Paradigm: sums $\rightarrow$ integrals, step payouts $\rightarrow$ smooth payouts.

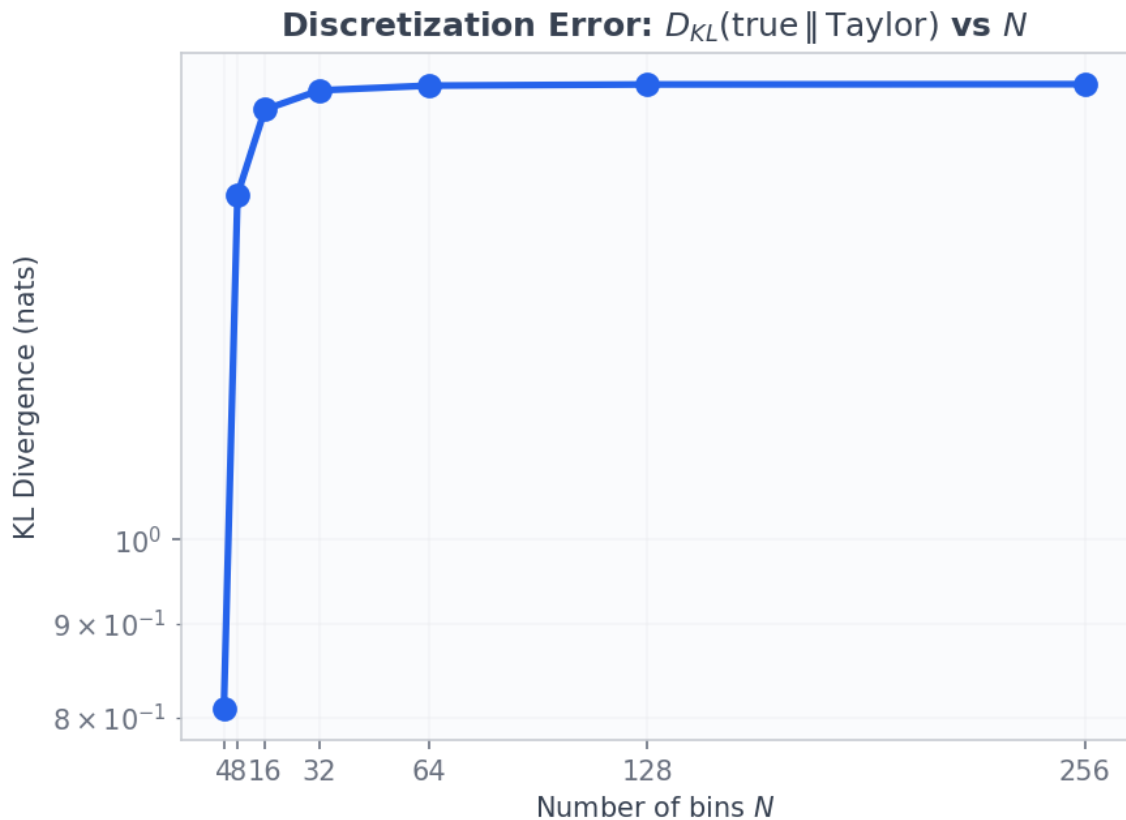


Figure 10: KL divergence between true and Taylor-approximated weights decreases with N .

12.5 Computational Efficiency

Paradigm (Normal case): A trade changes (μ, σ) . The AMM must verify L^2 norm, compute max-loss collateral (numerical optimization, no closed form), and verify $\max(f) \leq b$. DekantPM: $O(N)$ operations per trade — weights, XW , W^2 , reserve updates. For $N=256$, ~ 1000 arithmetic operations, feasible on Solana.

12.6 Where DekantPM Aligns

1. Same L_2 -norm invariant family.
2. Same Cauchy-Schwarz optimality: equilibrium $x \propto p$.
3. Same proportional LP model.
4. DekantPM converges to Paradigm as $N \rightarrow \infty$.

12.7 Where DekantPM Diverges

1. No separate backing constraint ($b = k$).
2. No smooth payout — step function per bin.
3. Restricted to Normal distributions for dist. trading.
4. Complete-set collateral (simpler but potentially less capital-efficient).
5. Quadratic probability extraction (distortion).

13. Summary of Mathematical Properties

1. Invariant preservation:

$$\sum_{i=1}^N x_i^2 = k^2 \quad (\text{within } \varepsilon = 256)$$

2. Solvency: vault $\geq k$, max redemption $\leq k$. Always solvent.
3. No arbitrage (within fees): probabilities sum to 1.
4. Market scoring rule: equilibrium $x \propto p$.

Constant	Value	Meaning
SCALE	10^9	Fixed-point denominator
INVARIANT_TOL	256	Max invariant drift
MAX_OUTCOMES	32	Max discrete outcomes
MAX_BINS	256	Max continuous bins
Z_CUTOFF	5	Gaussian tail cutoff
MIN_LIQUIDITY	10^6	1 USDC minimum
TRADE_FEE	30 bps	0.3% per trade
LP_FEE_SHARE	5000 bps	50% of fee to LPs
REDEMPTION_FEE	50 bps	0.5% on claims
CREATION_FEE	50 bps	0.5% on market creation

A. Appendix: Distribution Buy Quadratic

Seek $x'_j = x_j + \lambda \cdot W_j / W^2$ satisfying $\sum (x'_j)^2 = k'^2$. Expanding and solving the quadratic:

$$\sum \left(x_j + \frac{\lambda W_j}{W^2} \right)^2 = k'^2$$

$$\lambda^2 + 2 \sum W_j \cdot \lambda - W^2 \cdot \text{excess} = 0$$

$$\lambda = \sqrt{\sum W_j^2 + W^2 \cdot \text{excess}} - \sum W_j$$

B. Appendix: Integer Square Root

Newton's method for integer square root in u128:

$$x_0 = \lfloor n/2 \rfloor + 1$$

$$x_{t+1} = \left\lfloor \frac{x_t + \lfloor n/x_t \rfloor}{2} \right\rfloor$$

Stop when $x_{t+1} \geq x_t$. Converges in ≤ 64 iterations for u128.

C. Appendix: Glossary

Symbol	Meaning
N	Number of outcomes or bins
k	Total minted collateral
h_i, x_i	Reserve, position ($x_i = k - h_i$)
$\hat{p}_i$	Reported probability (x_i^2/k^2)
p_i	True probability ($x_i/\sum x_i$)
S	Scale factor (10^9)
W_j	Bin weight ($\sum W_j = S$)
μ, σ	Mean, std dev (trader's Normal)
XW, W^2	$\sum x_j W_j, \sum W_j^2$
λ	Distribution buy parameter
L	LP shares total
b	Backing amount (Paradigm only)